Hardware Recommendations for RooMonitor PCB

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VERSIONS CONTROL

Document version	Author	Modifications regarding previous version
1.0	Javier Longares Abaiz	First version

Summary

After several weeks working with RooMonitor boards, some issues have been found. The aim of this document is to guide the hardware engineers for fixing this issues in a new PCB version.

AC/DC STAGE

The power stage for AC-DC is now based in a Flyback converter. These converters are rarely used in general purpose electronic devices, and some of the reasons for avoiding them is that they produce really high current and voltages peaks, and that there is not isolation from AC and DC ground.

As a proposal to fit the box size and having something widely used and tested, as well as ready to pass the certifications, it is recommended to use the following module:

https://www.cui.com/product/resource/pbo-5-b.pdf

LINEAR REGULATOR

The DC circuit is being powered at 3.3V from linear regulator U6. However, we could face problems when the board is being powered from the battery and Vin is close to 3.4V, and considering that in Q1 we could have 0.1-0.2V of voltage drop, that limit could be quickly reached. The problem is that U6 has a dropout voltage of 0.15V, so when Vin is lower than 3.45V it will not power the DC circuit, and battery will looks as depleted, without being fully depleted.

We suggest to use a 3.0V low dropout voltage, that will the device lifetime longer when powered only from the battery.

BATTERY

- Q1 could be replaced by a Schotky diode. It is faster, smaller and cheaper.
- Voltage reading could be done without 2 batteries in paralel, for avoiding being continuously wasting the battery. An idea is to use a PNP transistor as serial switch, so we waste energy only when we are about to read the battery voltage

HARDWARE READY FOR LOW POWER MODE

Current design is not ready, at all, for low power mode. From the low power mode design point of view, there are some issues that should be taken in consideration for the future board:

- The microcontroller should be aware about the connection or disconnection to AC. Even though we are now measuring the Battery Voltage, that is not enough, since doing so, we enter in low power mode when the battery starts to be depleted. So we need a digital input in an ESP32's GPIO to determine that the AC is connected or not.
- There should be 2 linear regulators providing 3.0V: one for the microcontroller, and one for everything else in DC circuit. This second linear regulator should have an ENABLE pin, and this pin should be controlled to a GPIO on the microcontroller. The reason for this is that, with the current design, even if the microcontroller enters in low power mode, with a consumption of some milliseconds, the rest of DC circuit is wasting energy, so we are wasting unnecessarily the battery. With this modification, prior to enter in low power mode, the microcontroller will switch everything off first, and it will switch everything on again when waking up.

TAMPER CIRCUIT

If you are going to make a new board, it would be appreciate to include a tamper signal for detecting when someone opens the plastic enclosure.